In [2]:
```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns


pd.set_option('display.max_columns', None)
pd.set_option('display.width', 1000)

# Seed for reproducibility
np.random.seed(42)

# 1.Data Generation for 20 Members / Orders
n_rows = 20

order_ids = [f"ORD-{1000 + i}" for i in range(1, n_rows + 1)]
order_dates = (
    pd.date_range(start='2026-01-01', periods=n_rows, freq='D')
    .strftime('%Y-%m-%d')
    .tolist()
)
customer_ids = [f"CUST-{500 + i}" for i in range(1, n_rows + 1)]
customer_names = [
    'Aarav',
    'Priya',
    'Karthik',
    'Divya',
    'Rahul',
    'Ananya',
    'Vijay',
    'Sneha',
    'Arun',
    'Meena',
    'Sanjay',
    'Kavitha',
    'Surya',
    'Deepika',
    'Manoj',
    'Swathi',
    'Kumar',
    'Lakshmi',
    'Vignesh',
    'Ramya',
]
genders = np.random.choice(['Male', 'Female'], size=n_rows)
ages = np.random.randint(20, 55, size=n_rows)
cities = ['Chennai', 'Coimbatore', 'Madurai', 'Salem', 'Trichy']
city_list = np.random.choice(cities, size=n_rows)
states = ['Tamil Nadu'] * n_rows
regions = ['South'] * n_rows

product_ids = [f"PROD-{200 + i}" for i in range(1, n_rows + 1)]
product_names = [
    'Smartphone',
    'Laptop',
    'Headphones',
    'Smartwatch',
    'Tablet',
    'Bluetooth Speaker',
    'Keyboard',
    'Mouse',
    'Monitor',
    'Printer',
]
product_name_list = np.random.choice(product_names, size=n_rows)
categories = ['Electronics', 'Accessories']
category_list = np.random.choice(categories, size=n_rows)
brands = ['Dell', 'HP', 'Sony', 'Boat', 'Samsung', 'Apple', 'Lenovo']
brand_list = np.random.choice(brands, size=n_rows)

quantities = np.random.randint(1, 5, size=n_rows)
unit_prices = np.random.uniform(500, 50000, size=n_rows).round(2)
discounts = np.random.choice([0, 5, 10, 15, 20], size=n_rows)
cost_prices = (unit_prices * 0.7).round(2)

# Calculations
total_sales = (quantities * unit_prices * (1 - discounts / 100)).round(2)
profit = (total_sales - (quantities * cost_prices)).round(2)

payment_modes = ['UPI', 'Credit Card', 'Debit Card', 'Cash on Delivery']
payment_mode_list = np.random.choice(payment_modes, size=n_rows)
order_statuses = ['Delivered', 'Shipped', 'Processing']
order_status_list = np.random.choice(order_statuses, size=n_rows)
shipping_costs = np.random.choice([0, 50, 100, 150], size=n_rows)
delivery_days = np.random.randint(2, 7, size=n_rows)
ratings = np.random.randint(1, 6, size=n_rows)

# Create DataFrame
df = pd.DataFrame({
    'Order_ID': order_ids,
    'Order_Date': order_dates,
    'Customer_ID': customer_ids,
    'Customer_Name': customer_names,
    'Gender': genders,
    'Age': ages,
    'City': city_list,
    'State': states,
    'Region': regions,
    'Product_ID': product_ids,
    'Product_Name': product_name_list,
    'Category': category_list,
    'Brand': brand_list,
    'Quantity': quantities,
    'Unit_Price': unit_prices,
    'Discount_%': discounts,
    'Cost_Price': cost_prices,
    'Total_Sales': total_sales,
    'Profit': profit,
    'Payment_Mode': payment_mode_list,
    'Order_Status': order_status_list,
    'Shipping_Cost': shipping_costs,
    'Delivery_Days': delivery_days,
    'Rating': ratings,
})

print('--- (20 Members Dataset) ---')
display(df)

# 2. Visualizations (Bar Chart & Scatter Plot)
```

```
sns.set_theme(style='whitegrid')
plt.figure(figsize=(14, 6))

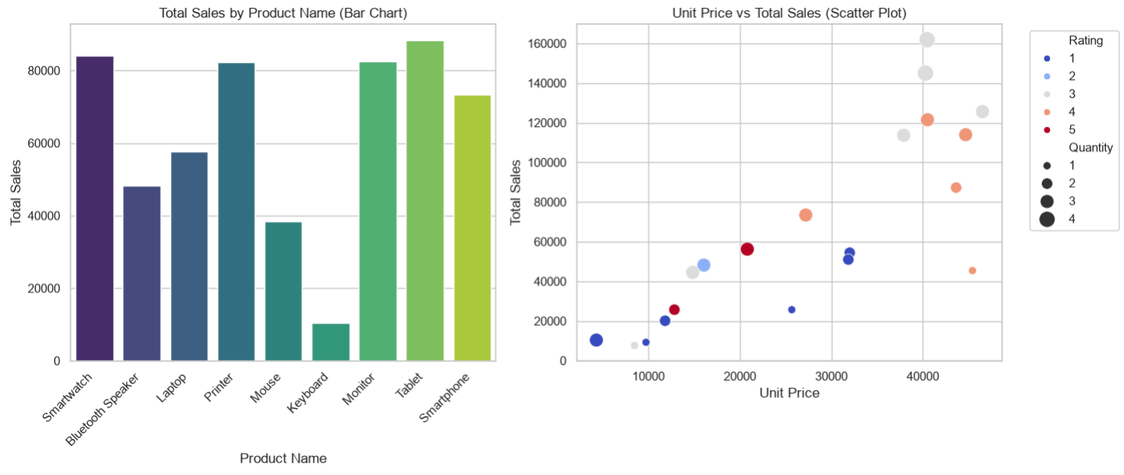
# Subplot 1: Bar Chart
plt.subplot(1, 2, 1)
sns.barplot(
    x='Product_Name',
    y='Total_Sales',
    data=df,
    hue='Product_Name',
    palette='viridis',
    legend=False,
    errorbar=None,
)
plt.title('Total Sales by Product Name (Bar Chart)')
plt.xticks(rotation=45, ha='right')
plt.xlabel('Product Name')
plt.ylabel('Total Sales')

# Subplot 2: Scatter Plot
plt.subplot(1, 2, 2)
sns.scatterplot(
    x='Unit_Price',
    y='Total_Sales',
    hue='Rating',
    size='Quantity',
    sizes=(50, 200),
    data=df,
    palette='coolwarm',
)
plt.title('Unit Price vs Total Sales (Scatter Plot)')
plt.xlabel('Unit Price')
plt.ylabel('Total Sales')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

plt.tight_layout()
plt.show()
```

```
--- (20 Members Dataset) ---
```

	Order_ID	Order_Date	Customer_ID	Customer_Name	Gender	Age	City	State	Region	Product_ID	Product_Name	Category	Brand	Quantity	Unit_Price	Discount_%	Cost_Price	Total_Sales	Profit	Payment_Mode	Order_Status	Shipping_Cost	Delivery_Days	Rating
0	ORD-1001	2026-01-01	CUST-501	Aarav	Male	41	Chennai	Tamil Nadu	South	PROD-201	Smartwatch	Electronics	Dell	2	32002.32	15	22401.62	54403.94	9600.70	Cash on Delivery	Shipped	50	4	1
1	ORD-1002	2026-01-02	CUST-502	Priya	Female	21	Salem	Tamil Nadu	South	PROD-202	Bluetooth Speaker	Electronics	Boat	3	16060.62	0	11242.43	48181.86	14454.57	UPI	Processing	50	4	2
2	ORD-1003	2026-01-03	CUST-503	Karthik	Male	43	Coimbatore	Tamil Nadu	South	PROD-203	Laptop	Accessories	Lenovo	1	25674.25	0	17971.97	25674.25	7702.28	Credit Card	Delivered	100	5	1
3	ORD-1004	2026-01-04	CUST-504	Divya	Male	49	Coimbatore	Tamil Nadu	South	PROD-204	Printer	Electronics	Dell	1	45424.54	0	31797.18	45424.54	13627.36	UPI	Delivered	100	5	4
4	ORD-1005	2026-01-05	CUST-505	Rahul	Male	21	Chennai	Tamil Nadu	South	PROD-205	Laptop	Accessories	Boat	2	12839.97	0	8987.98	25679.94	7703.98	Credit Card	Delivered	150	2	5
5	ORD-1006	2026-01-06	CUST-506	Ananya	Female	40	Coimbatore	Tamil Nadu	South	PROD-206	Printer	Accessories	HP	3	20813.95	10	14569.76	56197.67	12488.39	Cash on Delivery	Processing	150	2	5
6	ORD-1007	2026-01-07	CUST-507	Vijay	Male	52	Trichy	Tamil Nadu	South	PROD-207	Smartwatch	Accessories	Dell	3	37899.78	0	26529.85	113699.34	34109.79	Cash on Delivery	Shipped	0	3	3
7	ORD-1008	2026-01-08	CUST-508	Sneha	Male	31	Coimbatore	Tamil Nadu	South	PROD-208	Mouse	Electronics	Lenovo	2	11825.51	15	8277.86	20103.37	3547.65	Credit Card	Shipped	0	2	1
8	ORD-1009	2026-01-09	CUST-509	Arun	Male	41	Salem	Tamil Nadu	South	PROD-209	Keyboard	Electronics	Lenovo	3	4310.51	20	3017.36	10345.22	1293.14	Debit Card	Delivered	150	4	1
9	ORD-1010	2026-01-10	CUST-510	Meena	Female	44	Salem	Tamil Nadu	South	PROD-210	Monitor	Electronics	Apple	3	14842.70	0	10389.89	44528.10	13358.43	Credit Card	Shipped	100	5	3
10	ORD-1011	2026-01-11	CUST-511	Sanjay	Male	46	Salem	Tamil Nadu	South	PROD-211	Mouse	Electronics	Samsung	1	8480.45	10	5936.32	7632.41	1696.09	Debit Card	Shipped	50	2	3
11	ORD-1012	2026-01-12	CUST-512	Kavitha	Male	47	Salem	Tamil Nadu	South	PROD-212	Tablet	Accessories	Sony	3	46520.03	10	32564.02	125604.08	27912.02	UPI	Processing	150	2	3
12	ORD-1013	2026-01-13	CUST-513	Surya	Male	35	Trichy	Tamil Nadu	South	PROD-213	Laptop	Electronics	Boat	3	40501.96	0	28351.37	121505.88	36451.77	UPI	Processing	0	3	4
13	ORD-1014	2026-01-14	CUST-514	Deepika	Male	34	Madurai	Tamil Nadu	South	PROD-214	Tablet	Electronics	Apple	2	31853.49	20	22297.44	50965.58	6370.70	UPI	Processing	100	3	1
14	ORD-1015	2026-01-15	CUST-515	Manoj	Female	22	Chennai	Tamil Nadu	South	PROD-215	Mouse	Electronics	Sony	2	43637.30	0	30546.11	87274.60	26182.38	Cash on Delivery	Processing	150	4	4
15	ORD-1016	2026-01-16	CUST-516	Swathi	Male	26	Salem	Tamil Nadu	South	PROD-216	Printer	Electronics	Sony	4	40281.77	10	28197.24	145014.37	32225.41	UPI	Delivered	150	5	3
16	ORD-1017	2026-01-17	CUST-517	Kumar	Female	40	Coimbatore	Tamil Nadu	South	PROD-217	Monitor	Electronics	Dell	1	9735.22	5	6814.65	9248.46	2433.81	Debit Card	Processing	100	3	1
17	ORD-1018	2026-01-18	CUST-518	Lakshmi	Female	28	Salem	Tamil Nadu	South	PROD-218	Monitor	Accessories	Sony	3	44681.67	15	31277.17	113938.26	20106.75	UPI	Shipped	150	2	4
18	ORD-1019	2026-01-19	CUST-519	Vignesh	Female	37	Coimbatore	Tamil Nadu	South	PROD-219	Smartphone	Electronics	Samsung	3	27197.44	10	19038.21	73433.09	16318.46	Credit Card	Delivered	100	5	4
19	ORD-1020	2026-01-20	CUST-520	Ramya	Male	23	Coimbatore	Tamil Nadu	South	PROD-220	Monitor	Accessories	Lenovo	4	40468.29	0	28327.80	161873.16	48561.96	Credit Card	Shipped	50	5	3



In [ ]: